

Solution Requirements

SMARTBRIDGE / SKILL WALL

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Project: FitBuddy - AI Fitness Plan Generator using Gemini Models

Define Functional and Non-Functional Requirements

FR = what the system should do. NFR = how well the system should operate.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No login is required in the current MVP. The fitness-planning interface is directly accessible to users.	Low
2	Authorization levels	Single user role for the MVP. Users can enter fitness goals, preferences, and constraints and access the plan-generation features; no admin role is required.	Low
3	External interfaces	FastAPI REST endpoints connect the web interface with Google Gemini models to generate personalized workout plans from user-provided goals and preferences.	High
4	Transactions processing	The system validates a fitness request, gathers relevant profile context, sends a structured prompt to the selected Gemini model, and returns the generated plan.	High
5	Reporting	Present a readable plan with workout schedule, exercises, sets/repetitions or duration, progression guidance, and optional general nutrition suggestions. External export is not required in the MVP.	Medium
6	Business rules, etc.	Reject empty or invalid inputs. Use available goals, activity level, equipment, schedule, and limitations to tailor the result. Clearly label generated guidance as general information, not a medical diagnosis or substitute for professional advice.	High
7	Compliance to laws or regulations	Keep Gemini API credentials on the backend using environment configuration. Minimize collection and retention of personal and health-related information; do not expose secrets to the client.	High
8	Other	Provide clear validation and error messages, keep API and AI-generation logic modular, and expose core profile, plan-generation, and plan-history functions through REST APIs.	Medium

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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The health endpoint should respond quickly; plan generation should complete within normal model and network limits.	Health check < 1 second; normal AI requests complete without timeout.
2	Scalability	Keep FastAPI endpoints, profile/history handling, and AI-generation modules separated so API instances and model resources can be scaled independently.	API and AI-processing components can be scaled independently.
3	Security & Data Privacy	Keep Gemini API keys server-side and never expose secrets in HTML/JavaScript or source control. Validate and minimize sensitive user input; use only necessary profile context for generation.	No API secrets in frontend or repository; inputs validated; sensitive data minimized.
4	Reliability & Availability	Return clear validation errors and handle Gemini/API failures without crashing. Do not silently present failed or incomplete generation as a valid plan.	Invalid input returns 4xx; service/model failures return controlled errors.
5	Usability & Accessibility	Provide a simple responsive interface with clear goal and preference inputs, readable plan output, and accessible labels and contrast.	Core inputs and generated plan are easy to locate and read on desktop and mobile.
6	Other	Use modular Python files and automated tests for core API, validation, and plan-generation behavior.	Core modules remain independently testable; tests run before submission.